



Machine learning-based prediction of non-aeration linear alkylbenzene sulfonate mineralization in an oxygenic microalgal-bacteria biofilm

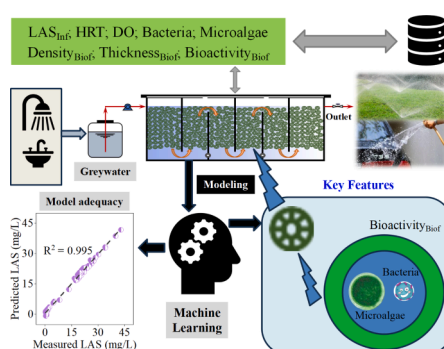
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HIGHLIGHTS

- The support vector machine model is capable to predict linear alkylbenzene sulfonate removal.
- Variable importance, PDP and SAE were applied for model explanation and outputs responding.
- The average R^2 , RMSE, and MAE were 0.961, 0.251, and 0.153 for testing set, respectively.
- The optimum operation conditions of MBBfR for greywater treatment were obtained.
- Machine learning model achieved the reduction of treatment time and energy-supply.

GRAPHICAL ABSTRACT



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ABSTRACT

Microalgal-bacteria biofilm shows great potential in low-cost greywater treatment. Accurately predicting treated greywater quality is of great significance for water reuse. In this work, machine learning models were developed for simulating and predicting linear alkylbenzene sulfonate (LAS) removal using 152-days collected data from a baffled oxygenic microalgal-bacteria biofilm reactor (MBBfR). By using nine variables including influent LAS, hydraulic retention time (HRT), biofilm density and thickness, specific oxygen production and consumption rates, microalgae and bacteria concentrations, and dissolved oxygen (DO), the support vector machine (SVM) model enabled the accurate LAS removal prediction (training set: $R^2 = 0.995$, (root mean square error, RMSE) = 0.076, (mean absolute error, MAE) = 0.069; testing set: $R^2 = 0.961$, RMSE = 0.251, MAE = 0.153). SVM can be also successfully applied for MBBfR operation optimization (HRT = 4.28 h, DO = 0.25 mg/L) that achieving accurate prediction of LAS mineralization.

1. Introduction

Greywater is generated from domestic sources including bathing and laundry room, which contains higher water volume but lower levels of

organics and nutrients compare with blackwater in domestic wastewater (Friedler, 2004; Shaikh et al., 2019). The reclamation of treated greywater has revived great attentions due to energy-efficient treatment technologies, low water supply cost, and reducing the load on

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centralized wastewater treatment plant (Wu, 2018). However, greywater also contains large amount of linear alkylbenzene sulfonate (LAS) that contributing to 40–80 % of total organics (Luo et al., 2023; Zhou et al., 2022), which shows high risk to human and toxicity to biological greywater treatment systems (Cui et al., 2022b; Wang et al., 2022; Zhou et al., 2020a). LAS can be rapidly and completely biodegraded (>98 %) by aerobic biological treatment methods (Babaei et al., 2019; Zhou et al., 2020b), but which requires large amounts of oxygen (O_2) and exhibits high processing cost. Data-driven decentralized building-based water technology has been applied to separate greywater, rainwater and wastewater, and treat greywater before discharged into sewage pipes (Li et al., 2022).

As a novel biological technology, microalgal-bacteria biofilm reactor (MBBfR) demonstrates significant advantages in greywater treatment,

including efficient multiple pollutants removal, low energy demand due to aeration-free operation, reduced greenhouse gas emission, and considerable potential for resource utilization (Chen et al., 2013; Fardi et al., 2024; Schlafer and Meyer, 2017). Various operation parameters and environment factors, including hydraulic retention time (HRT), biofilm physicochemical properties, dissolved oxygen (DO) concentration, system pH value and light intensity, will impact pollutants removal efficiency and biomass growth rate in the MBBfR (Igou et al., 2023; Quijano et al., 2017). However, the complex greywater characteristics, variable environment and operation conditions will lead to the collapse of microalgal-bacteria biofilm (Quijano et al., 2017; Yin et al., 2023). Operation conditions optimization and pollutants removal prediction, especially for LAS mineralization during greywater treatment, should be important to achieve the practical application of MBBfR in low-energy-

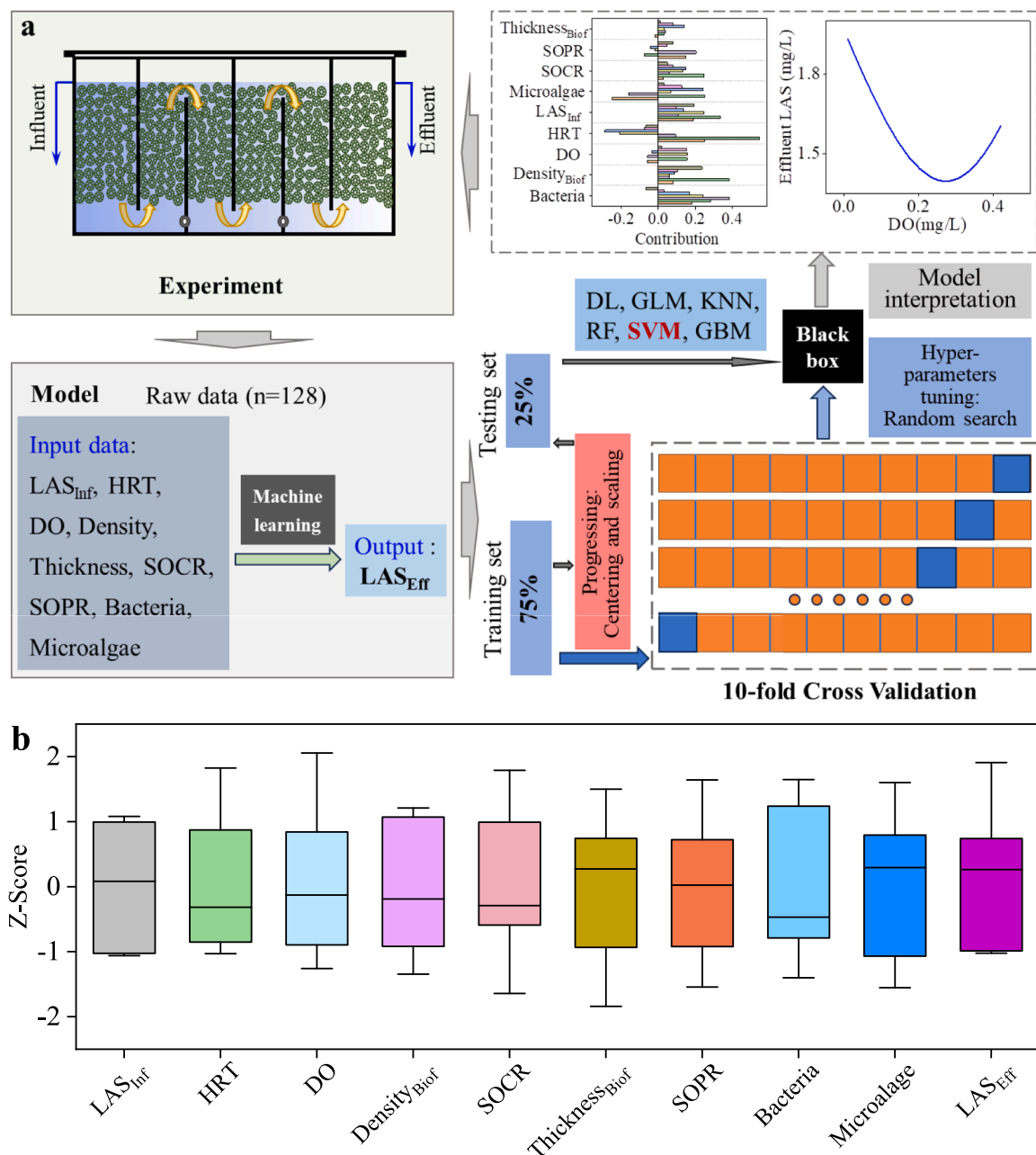


Fig. 1. (a) Schematic of the workflow in this work and (b) the boxplot of normalized original dataset. The experiment design, the process of machine learning modeling and the model interpretation are described respectively. LAS_{Inf} and LAS_{Eff} are influent and effluent linear alkylbenzene sulfonate (LAS) concentrations; Density_{Biof} and Thickness_{Biof} are biofilm density and thickness, respectively; SOPR and SOCR are specific oxygen production and consumption rates, respectively; HRT is hydraulic retention time and DO is dissolved oxygen.

input greywater treatment and reuse.

During LAS mineralization in greywater treatment, understanding the change of mechanism-based LAS removal should be important for the maintaining of stable MBBfR (Baeten et al., 2019). However, first principles is inapplicable for the modeling of exhibiting nonlinear behavior (Newhart et al., 2019). Big data-based machine learning (ML) models become a powerful tool for revealing unknown rules (Zhong et al., 2021), which enabled a highly intelligent and autonomous greywater treatment system (Fu et al., 2022). The dominant ML algorithm based on laboratory obtained data always contain artificial neural network (Gong et al., 2019), support vector machine (Yasmin et al., 2020), deep learning (Shi and Xu, 2018), and random forests (Wang et al., 2021). Deep learning is suitable for complex data sets, random forest is ensemble learning algorithms, and the support vector machine applies to small samples and nonlinear regression. ML is normally used in the prediction of sewage treatment effluent quality (Harrou et al., 2018), gas production (Sadoune et al., 2023), anomaly monitoring (Thompson and Dickenson, 2021) and water quality in natural bodies by considering the complex interactions among related variables (Chen et al., 2023). In the process of wastewater treatment, ML enabled the prediction of water quality and the optimization of process operation conditions (Zhong et al., 2021). Reaction principles should be also combined with data drive during ML-based prediction and optimization (Corominas et al., 2018). Deep learning network enabled the prediction of reactor performance and the optimization of operation conditions in the biofilm-based wastewater treatment system (Chang et al., 2022; Shi and Xu, 2018). In the membrane bioreactor, ML can exhibit the non-linear relationship between microorganisms and environmental conditions, as well as the prediction of transmembrane pressures in various operation cycles, which could not be detected by traditional methods (Wijaya and Oh, 2023). Despite the extensive application of ML in various sewage treatment systems, ML-based LAS mineralization prediction and process optimization during greywater treatment in the MBBfR is still limited. ML serves as a valuable tool in environmental science and engineering. Model interpretation methods, including variable importance analysis, Shapley additive explanations (SHAP), and partial dependence plots (PDP), are commonly employed (Zhong et al., 2021). The SHAP value quantifies the positive and negative contributions of the reaction conditions to the outcomes (Jeong et al., 2023; Yang et al., 2024), while the PDP illustrates how the predicted results change with respect to the variables (Xu et al., 2023).

In this work, ML was applied for MBBfR with limited available datasets. Original data were pre-processed and cleaned, and the variance inflation factor –based feature-selection was applied for reducing multicollinearity. By using the influent greywater quality parameters and biofilm characteristics, six ML models were used for the simulation and prediction of effluent LAS concentration. The selected optimal ML model was interpreted to illustrate the effect of biofilm and influent greywater characteristics on effluent LAS, and the optimization of MBBfR operation conditions. The purpose of this model is to provide a powerful and reliable tool that covering the gap to connect biofilm and influent characteristics with LAS mineralization efficiency directly, as well as enabling process optimization of MBBfR for energy-efficient greywater treatment and reuse.

2. Methods and materials

2.1. Experimental design and raw data collection

Laboratory MBBfR (Fig. S1) was set up according to previous published works (Chen et al., 2013; Schlafer and Meyer, 2017). Activated sludge and *Synechocystis* sp. PCC6803 were inoculated into the system with biomass concentration of 1.19 ± 0.04 g total suspended solids/L and bacteria: microalgae mass ratio of 1:1. Greywater was prepared according to the National Sanitation Foundation/American National Standards Institute (ANSI) Standard 350 (NSF/ANSI 350, 2011) and the

detailed information of the influent greywater characteristics (see Supplementary material). MBBfR was continuously operated for 152 days at room temperature (23.1 ± 0.4 °C) with the hydraulic retention times (HRT) of 28 h, 20 h, 14 h, 10 h, 6 h, 4 h and 5 h at each stage. Effluent liquid sample was taken every three days and biofilm samples were collected at the end of each stage with stable reactor performance. As the selected variables for the simulating and predicting using ML models (totally 128 data sets), we recorded or measured reactor HRT, dissolved oxygen (DO), influent and effluent LAS concentrations, biofilm density (Density_{Biof}) and thickness (Thickness_{Biof}), biofilm specific oxygen production (SOPR) and consumption rates (SOCR), biomass concentrations of microalgae and bacteria.

2.2. Data processing

For the training of ML algorithms, cleaning process of the training datasets was necessary, which normally includes removing outliers, filling missing data, randomizing the dataset and normalizing all data features to the same scale. Outliers in the target datasets were deleted during data collection and linear interpolation was applied for missing points filling (van Buuren and Groothuis-Oudshoorn, 2011). During the splitting of training and testing datasets, randomization was applied for removing the impact of staged operation and various reactor sampling points (Willett, 1999). In this work, the datasets collected from the MBBfR are randomly split into training dataset (75 %) and testing dataset (25 %), which exhibited the similar statistical properties that ensuring the proper model evaluation. A genetic algorithm is employed for feature selection, enabling the identification of essential variables. To reduce the meaningless simulating and improve the model calculation accuracy that contributing from the potential linear correlations between the collected datasets (Zhang et al., 2022), normalization method was applied to decrease the impact between all target input variables (see Supplementary material).

Data standardization conforms to normal distribution and reduces the difference between data ranges, which facilitates the construction of ML model in subsequent steps. Fig. 1b depicts a boxplot of nine input characteristics and effluent LAS from the original data set, which are standardized and preprocessed. The mean and standard deviation of the raw data was shown in the Supplementary material. During model testing process, the data is standardized using the training dataset to minimize data leakage.

2.3. Model development and evaluation

The ML algorithms used in this work include generalize linear model (GLM), deep learning (DL), k-nearest neighbor (KNN), support vector machine (SVM), random forest (RF), and gradient boosting machine (GBM), all of which are normally applied for the simulating, predicting and process optimization in various biological wastewater treatment systems (Xia et al., 2024; Zhong et al., 2021). Structural risk minimization-based SVM, according to statistical learning theory, was normally applied for minimizing modelling error and preserving model simplicity. In SVM, a boundary consists of a center line and two parallel boundary lines and the core principle of SVM regression is to fit the data by minimizing the prediction error while maintaining a boundary in the fitting process (Smola and Schölkopf, 2004). RF employs an ensemble learning method composed of many decision trees, with the output being a summary of the best solutions to the problem (Breiman, 2001). KNN indicates that if most of the k nearest samples in the feature space belong to a certain category, the sample is classified into that category (Samworth, 2012). DL is a deep neural network that mimics how the human brain processes information and each neuron is connected to other neurons and has an associated weight and threshold (Schmidhuber, 2015). GBM linearly combines base learners with different weights to reuse the excellent learners. The gradient is raised to produce a highly robust and interpretable model for regression,

Table 1
RMSE, MAE and R^2 of six different machine learning methods.

Model	RMSE		MAE		R^2	
	Training	Testing	Training	Testing	Training	Testing
DL	0.424	0.626	0.271	0.350	0.839	0.765
GLM	0.279	0.523	0.192	0.337	0.921	0.818
KNN	0.068	0.416	0.028	0.187	0.995	0.894
RF	0.135	0.387	0.071	0.202	0.982	0.908
SVM	0.076	0.251	0.069	0.153	0.995	0.961
GBM	0.001	0.324	0.001	0.198	0.999	0.933

RMSE is root mean square error, MAE is mean absolute error, DL is deep learning, GLM is generalized linear model, KNN is K-nearest neighbor, RF is random forest, SVM is support vector machines and GBM is Gradient boosting machine.

particularly suited for mining less-than-clean data (Friedman, 2001). Data processing and model development were achieved using the *caret* package (Kuhn, 2008). During model development and evaluation, determination coefficient (R^2), root mean square error (RMSE) and mean absolute error (MAE), were utilized for the assessing of modeling performance on both training and testing dataset (see Supplementary material). High value of R^2 and low values of both RMSE and MAE indicated excellent model performance (Palansooriya et al., 2022). Cross-validation was used in model development that improving prediction accuracy and model generalization capacities (Li et al., 2020).

2.4. Model interpretation

The ML model only considers the relationship between input and

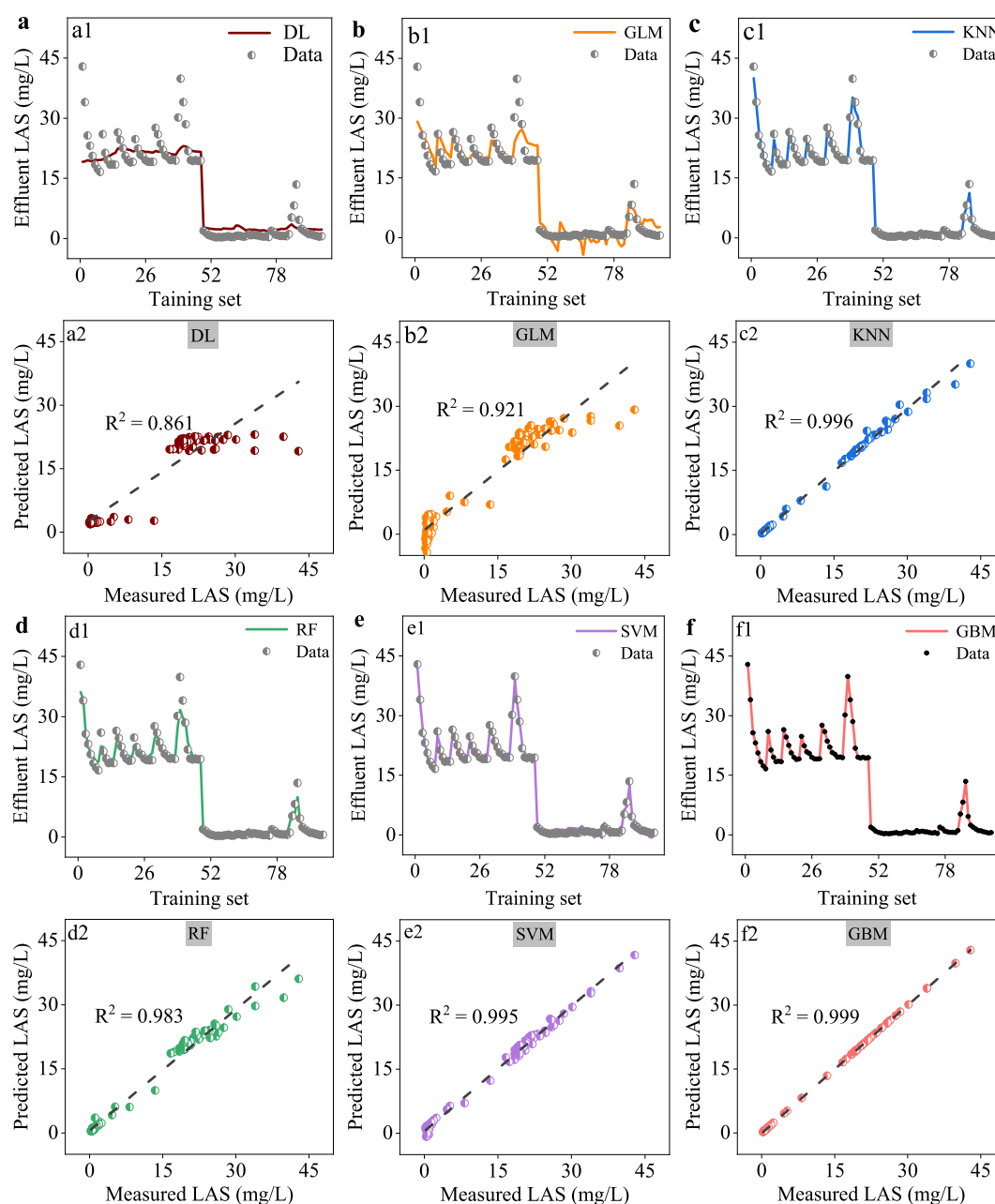


Fig. 2. Comparison between predicted and actual values (training set) of the linear alkylbenzene sulfonate (LAS) mineralization in the microalgal-bacteria biofilm reactor by different machine learning algorithms (a1, b1, c1, d1, e1, f1). Six different models predict performance in the training set (a2, b2, c2, d2, e2, f2). (a) Deep learning (DL), (b) Generalized linear model (GLM), (c) K-nearest neighbor (KNN), (d) Random forest (RF), (e) Support vector machines (SVM), (f) Gradient boosting machine (GBM).

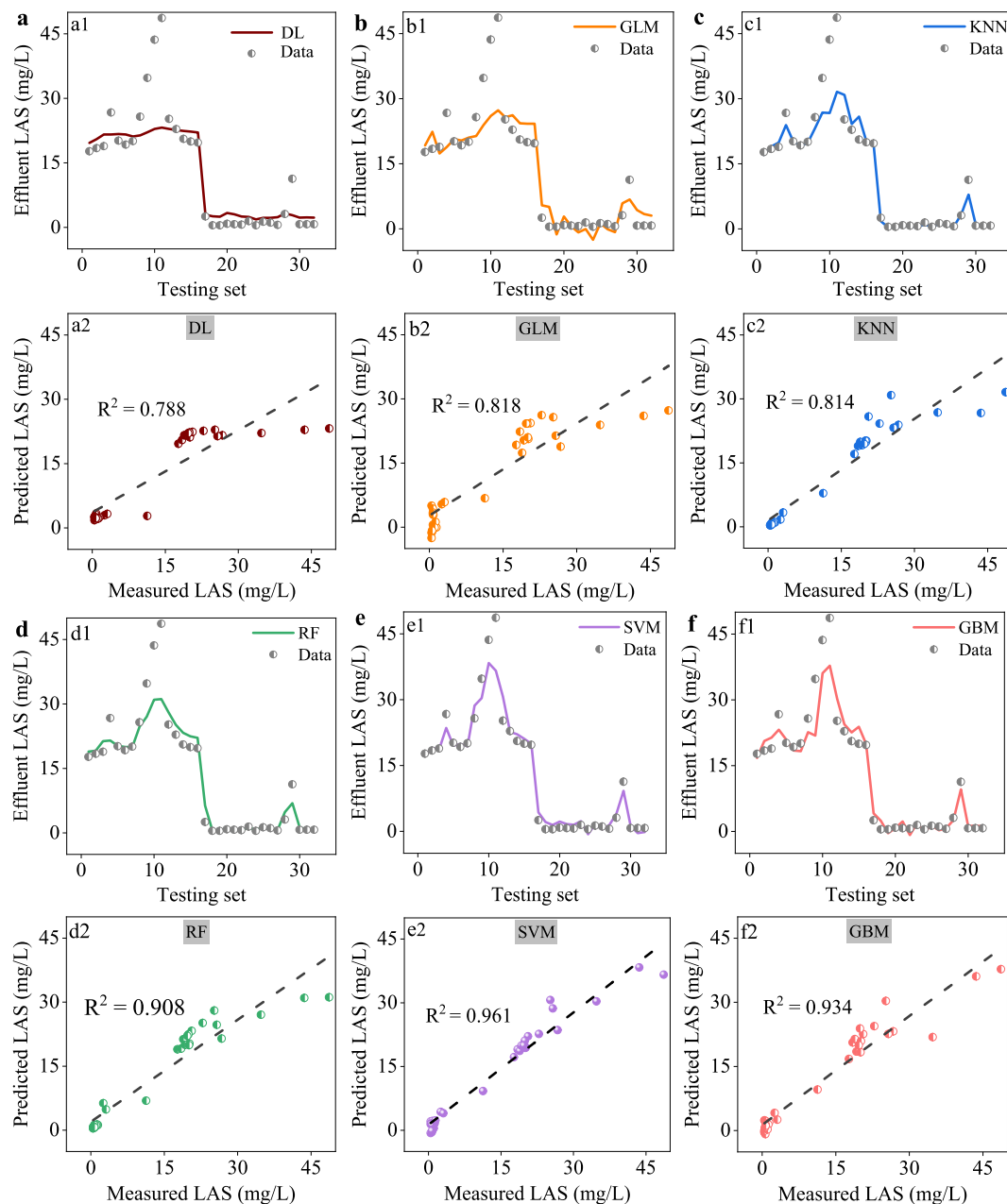


Fig. 3. Comparison between predicted and actual values (testing set) of the linear alkylbenzene sulfonate (LAS) mineralization in the microalgal-bacteria biofilm reactor by different machine learning algorithms (a1, b1, c1, d1, e1, f1). Six different models predict performance in the testing set (a2, b2, c2, d2, e2, f2). (a) Deep learning (DL), (b) Generalized linear model (GLM), (c) K-nearest neighbor (KNN), (d) Random forest (RF), (e) Support vector machines (SVM), (f) Gradient boosting machine (GBM).

output features, which enabled fuzzy rules accurate prediction based on the laboratory obtained data (Liao et al., 2023). For practical applications, variable importance analysis, partial dependence plots (PDP) (Greenwell, 2017) and Shapley additive explanations (SHAP) (Štrumbelj and Kononenko, 2014) are normally applied for incorporating of the best model interpretation from different viewpoints. Detailed information about the mentioned technologies were provided (see Supplementary material). The variable importance indicates the relative importance of the most significant model variables (Fisher et al., 2018), and PDP provide a visual representation of the marginal effect of a variable on the response. The impact of a variable is measured by changing the mean response and PDP assumes independence among the feature during PDP computing process (Friedman, 2001). The SHAP summary plot illustrates the contribution of each feature, and the sum of

these feature contributions, along with the bias term, equals to the raw model prediction prior to applying the inverse link function (Štrumbelj and Kononenko, 2014). Model explanations were generated using the DALEX package (Biecek, 2018), and all R codes developed for this work is available at <https://github.com/xialibo96/LASdata>.

3. Results and discussion

3.1. Model development and performance

Data standardization and normalization enabled the same distribution in both the training and testing datasets. The 10-fold cross-validation method was applied for model training and hyperparameters adjusting in the training set. According to the model

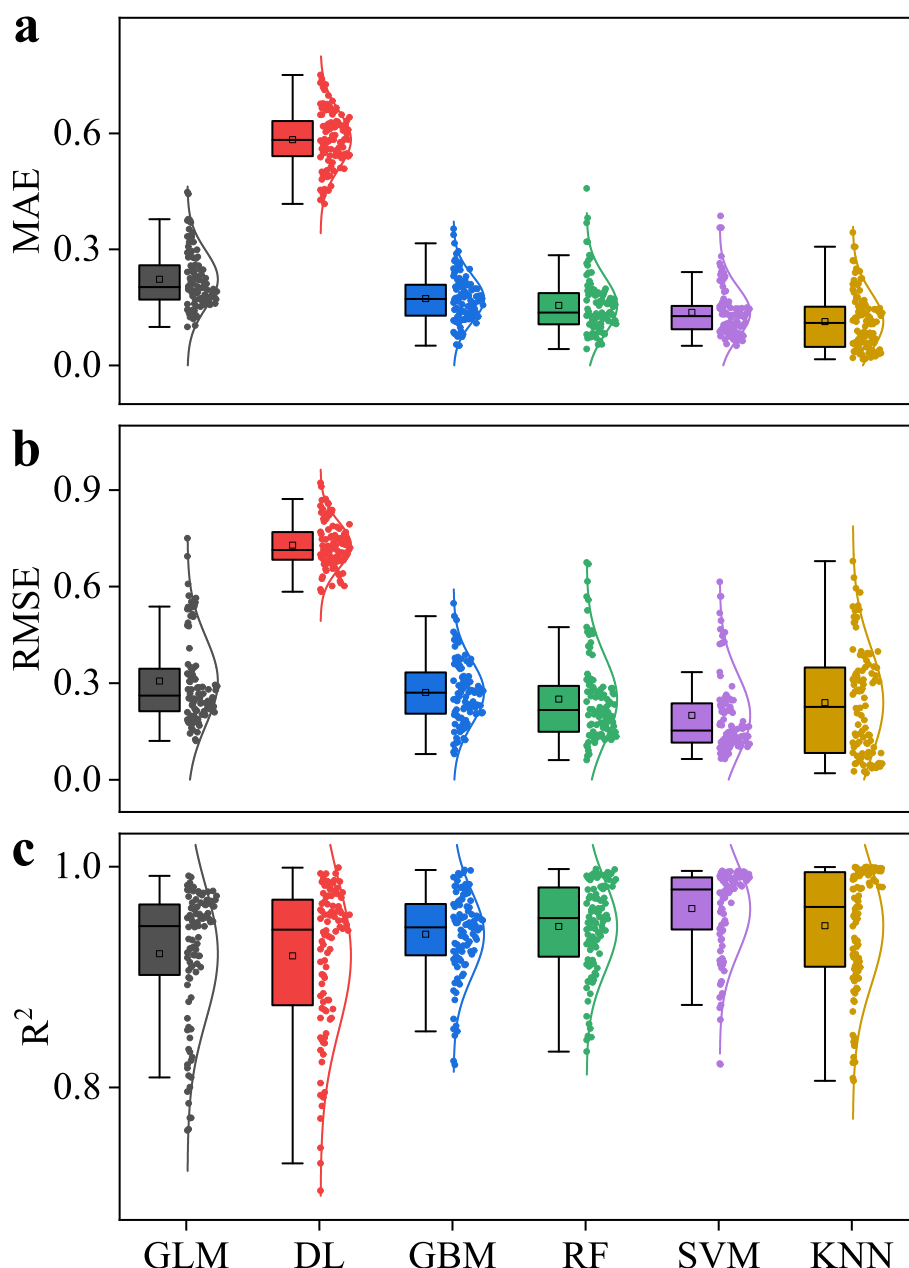


Fig. 4. The box plots shows the performance comparison between GLM, DL, GBM, RF, SVM, KNN using resampling based on the training data set. MAE: Mean absolute error, RMSE: Root mean square error and R^2 : Determination coefficient. GLM: Generalized linear model GLM, DL: Deep learning, GBM: Gradient boosting machine, RF: Random forest, SVM: Support vector machines and KNN: K-nearest neighbor.

evaluation criteria, all algorithms were validated using a 10-fold cross validation. The training set data was divided into 10 subsets on average, with 9 subsets used for training and 1 subset used for validation. Each calculation was repeated 10 times using a random search to optimize the hyperparameters. The optimal hyperparameters were determined by averaging the results from these 10 iterations (Efron, 1983). Since model hyperparameters significantly affect ML performance, both model selection and related hyperparameters adjustments should be conducted simultaneously during ML development. Random search was employed to select the optimal hyperparameters after 10-fold cross-validation, and the final model of different algorithms was determined by the lowest RMSE. Their ratings in the training set are shown in Table 1, which displays the related ratings in the training set.

In this work, six ML algorithms were selected for models building. After hyperparameter tuning, six different models with the related optimal hyperparameters were obtained, representing the best

performance for each algorithm. As shown in Fig. 2, the simulated values of the target models were compared with the actual values and related correlation coefficients in the training dataset. In the training set, The R^2 values for the algorithms in the training dataset were as follows: DL (0.839), GLM (0.921), KNN (0.995), RF (0.982), SVM (0.995), and GBM (0.999). According to the no free lunch theorem (Wolpert and Macready, 1997), no general learning algorithm can achieve good performance across a variety of tasks; therefore, specific analyses of particular problems are necessary. The SVM algorithm is particularly effective for small sample data, and the hyperparameter adjustment is relatively straightforward. The predictive performance of ML on membrane bioreactor was reported as 97 % (Chang et al., 2022), 95.7 % (Zaghloul et al., 2021), and over 91.6 % (Wijaya and Oh, 2023). The accuracy of the flow and energy consumption models exceeded 90 % (Kusiak et al., 2013). All above reported results were lower than the predicted performance according to the target ML models in this work.

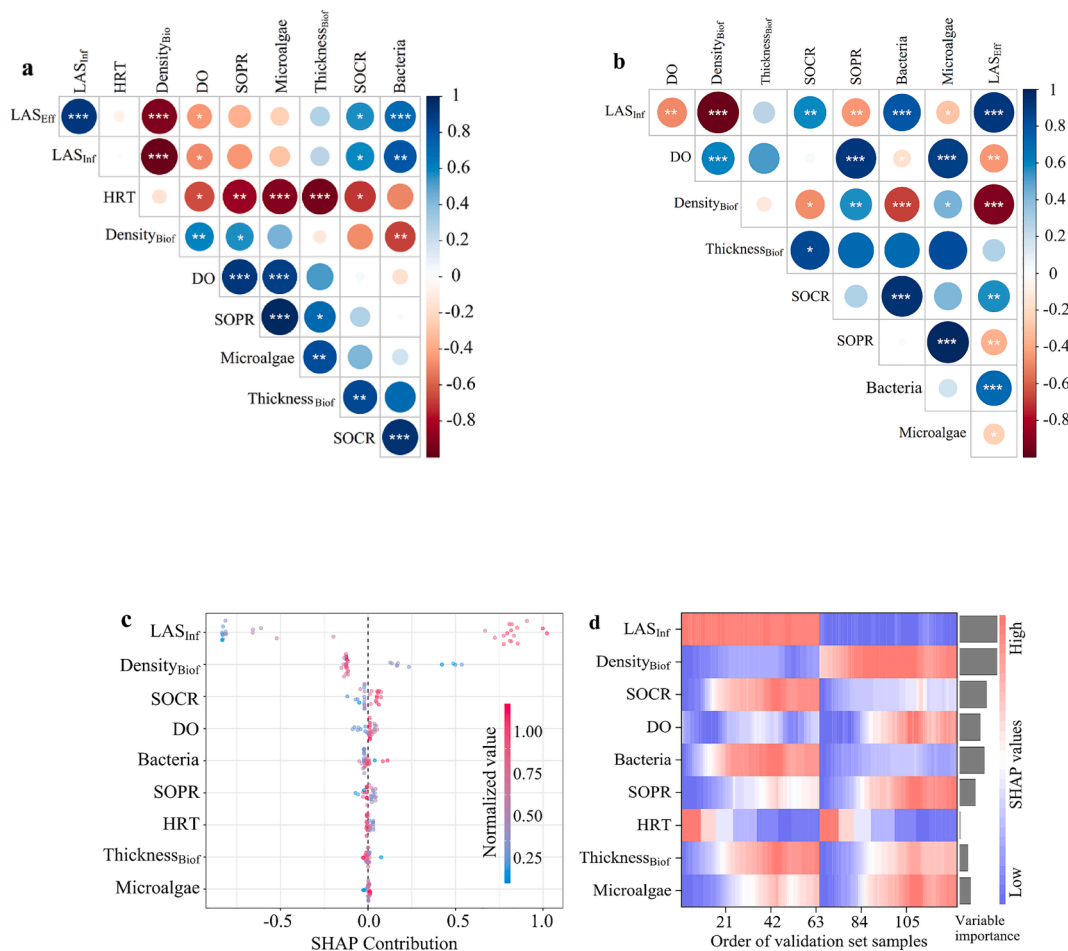


Fig. 5. The heat map shows the degree of correlation between the original dimensional characteristics of the linear alkylbenzene sulfonate (LAS) mineralization in the microalgal-bacteria biofilm reactor. (a) Features correlations between operation conditions, biofilm characteristics and LAS concentrations, (b) Features correlations after automated construction and culling of features using the support vector machines (SVM) model, (c) Global interpretation about distinguishing feature values and (d) supervised clustering of testing set samples with variable importance of related characteristics. LAS_{Eff} and LAS_{Inf} are effluent and influent LAS concentrations, respectively; Density_{Biof} and Thickness_{Biof} are biofilm density and thickness, respectively; SOPR and SOCR are specific oxygen production and consumption rates, respectively; HRT is hydraulic retention time and DO is dissolved oxygen. One star denotes significance $P < 0.05$, two stars denote significance $P < 0.01$, and three denote significance $P < 0.001$.

Fig. 2 compares the predicted and actual values of the six models in the training set. The performance of both DL and GLM is poor in the training set. The other four machine learning algorithms all performed well ($R^2 > 0.98$). After model training, the optimal models for six different ML algorithms were obtained and Table 1 shows the RMSE, MAE and R^2 values in both training and testing datasets. Fig. 3 compares the predicted and actual values of the six models in the testing set. During the model training process, the performance was optimized by adjusting the hyperparameters according to the characteristics of the training set data. However, the data from the testing set needs to be validated before practical application. KNN ($R^2 = 0.814$) and RF ($R^2 = 0.908$) performed poorly in the testing set. The resampling distribution of the model is used to represent the differences between the models (Eugster and Leisch, 2011). The resampling data of MAE, RMSE and R^2 for the six models are compared, and the optimal model is SVM (Fig. 4). By comparing the RMSE of the models, SVM was confirmed as the optimal model as which exhibited lower errors in both training and testing datasets. The RMSE, MAE and R^2 values for SVM were 0.251, 0.153 and 0.961, respectively. Although the GBM showed the smallest error in the training dataset ($R^2 = 0.999$), its performance in the testing dataset was poor, which should be contributed to overfitting (Cawley and Talbot, 2010). The GBM is an ensemble model. When the hyperparameters of the model are adjusted through cross validation, the

results are averaged, which is similar to the ensemble process and therefore lead to overfitting. Because of the small dataset size and the simplicity of the input features, the latest deep learning algorithms could not achieve optimal results. Consistent with previous findings that the optimal ML algorithm applied in membrane biofilm reactors for wastewater treatment is SVM (Harrou et al., 2018; Yasmin et al., 2019). As ML models are statistical and data-driven rather than based on reaction mechanisms, small datasets could limit its applicability.

3.2. Model interpretability

3.2.1. Correlation analysis of different variables to LAS removal

As the obtained optimal model, SVM can be also applied for the prediction of effluent LAS in the MBBfR with different variables. When applying the black box model to specific problems, the interpretability of the optimal ML model should be crucial. Fig. 5a exhibits the degree of correlation heatmap between the original dimensional characteristics of LAS mineralization performance and input variables in the MBBfR. Density_{Biof} showed negative correlation with LAS_{Eff} ($R^2 = -0.911$, $p < 0.001$). Consistent with previous findings that high microalgal-bacteria biofilm density favored LAS removal (Chen et al., 2013; Schlafer and Meyer, 2017). But much higher Density_{Biof} inhibited the transfer of nutrients and lights, leading to the reduced pollutants removal efficiency

(Sanchez-Huerta et al., 2022). Thus, regulation of biofilm density should be an effective strategy for maintaining stable biofilm function and pollutants removal. LAS_{Inf} exhibited positive relationship with LAS_{Eff} ($R^2 = 0.916$, $p < 0.001$). Higher LAS_{Inf} led to the increased LAS_{Eff} in the MBBfR due to the rose of LAS loading (Yin et al., 2023). Additionally, biotoxicity of LAS also led to the reduction of biomass concentration, bioactivity and enzymes functional for LAS mineralization in the biofilm (Zhou et al., 2020a; Zhou et al., 2021; Zhou et al., 2020b), resulting to the decreased organics removal efficiency and increased LAS_{Eff} . Bacteria concentration in the symbiotic biofilm showed positive correlation with LAS_{Eff} ($R^2 = 0.701$, $p < 0.001$). Microalgae can be decomposed by bacteria (Li et al., 2023; Wei et al., 2023), low microalgae and high bacteria concentrations led to insufficient oxygen in the MBBfR (Yin et al., 2023), leading to the inefficient LAS biodegradation and high LAS_{Eff} . HRT ($R^2 = -0.072$, $p = 0.79$) and Microalgae ($R^2 = -0.241$, $p = 0.18$) showed weak correlations with effluent LAS. However, a change in HRT induced internal changes in the biofilm within the MBBfR during the experimental operation stage. HRT exhibited a strong correlation with the biofilm thickness ($R^2 = -0.941$, $p < 0.001$) and microalgae ($R^2 = -0.902$, $p < 0.001$), which in turn impact LAS removal in the system. Fig. 5b presents a heatmap illustrating the correlation between microalgal-bacteria biofilm characteristics and operating conditions. There is no multicollinearity among these variables, all of which can be utilized as input variables for the model.

3.2.2. SHAP contribution value of different variables to LAS removal

SHAP summary plot (Fig. 5c) illustrates the contribution of each input variable. The horizontal axis describes the distribution of the influence of individual SHAP values on the model output. The samples are stacked vertically, with larger eigenvalues represented by red. The LAS_{Inf} and SOCR exhibit positive effects that hinder pollutants removal. Specifically, SOCR consumes oxygen in the reactor, which is detrimental to LAS mineralization under aerobic conditions. Conversely, $Density_{Biof}$ exhibits a negative effect and facilitates for pollutants removal. As shown in Fig. 5d, the variable importance of the optimal model was consistent with the correlation, indicating an inherent mechanism among the input features of the SVM model. The global importance of $Density_{Biof}$, which ranked second, is 99.03 % of LAS_{Inf} , while the importance of SOCR, ranked third, is 71.58 % of LAS_{Inf} . The predictive variables are crucial for model construction and the data-driven model can better capture the mechanism in certain cases (Fisher et al., 2018). After model interpretation, it is found that the variable importance is significant even with weak correlation. Consistent with previous findings that industrial wastewater sources serve as a reliable predictor despite their weak association in the analysis (Liu et al., 2023).

SHAP method was used to reveal the influence of various ions and membrane properties on the model prediction (Jeong et al., 2023). And it also reveals the influence of pollutant properties and reaction conditions (Sun et al., 2023) and finds key input characteristics of the osmotic membrane bioreactor (Chang et al., 2022). The impact of SHAP values on bio-oil composition, influenced by input characteristics (Chen et al., 2024) and reaction temperature, significantly contributes to the conversion of tar (Shafizadeh et al., 2023). The SHAP contribution values indicate that the contribution of LAS_{Inf} is scattered and deviated from the center, highlighting its substantial influence on LAS mineralization (Fig. 5d). High influent concentrations of LAS can adversely affect the overall reactor operation performance and further hinder pollutants removal. The impacts of biofilm density on LAS_{Eff} contrasts with that of LAS_{Inf} . According to SHAP results, the contributions of other input variables to LAS removal are not significant. Because of the toxicity of LAS and changes in biofilm physicochemical properties, these input characteristics have varying contribution values in the first and third regions of the MBBfR (Fig. 5d).

The SHAP local explanation plot illustrates the contribution of features to HRT variations (Štrumbelj and Kononenko, 2014). The sum of the feature contributions and the bias term equals to the raw prediction

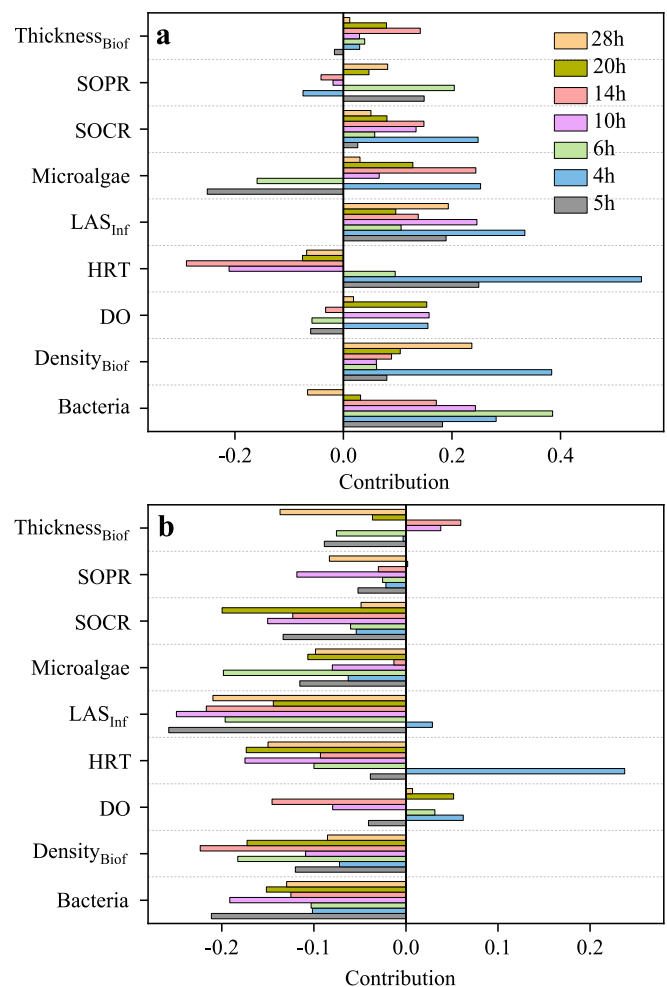


Fig. 6. SHAP local explanation plots for the linear alkylbenzene sulfonate (LAS) mineralization in the microalgal-bacteria biofilm reactor. The first (a) and third (b) operating regions of the reactor, the contribute values of input variables in seven different HRT. LAS_{Inf} is influent LAS concentration; $Density_{Biof}$ and $Thickness_{Biof}$ are biofilm density and thickness, respectively; SOPR and SOCR are specific oxygen production and consumption rates, respectively; HRT is hydraulic retention time and DO is dissolved oxygen.

of the model (Štrumbelj and Kononenko, 2010). Fig. 6 depicted the SHAP local explanation plots for the LAS mineralization in the first and third operation regions of the battled MBBfR. In the first region, the system generally showed poor pollutants removal efficiency. In the third region of MBBfR (Fig. 6b), input variables contributed negatively to effluent LAS. After continuous adsorption and biodegradation of LAS in the first region of MBBfR, the decreased LAS concentration led to the reduction of LAS biotoxicity, enabling stable and efficient LAS removal in the third region of the system (Cui et al., 2022a; Cui et al., 2022b). Longer HRT and high microalgae concentration led to relatively low effluent LAS, but other influent parameters and biofilm physicochemical properties positively impact effluent LAS. Results indicated that MBBfR enabled the efficient removal of LAS, which should be contributed to biomass biosorption (bacteria and microalgae) and biodegradation (dominant by bacteria) (Chen et al., 2013; Li et al., 2023). Generally, longer HRT (≥ 5 h) enabled the stable and efficient LAS mineralization with low LAS_{Eff} , lower HRT (≤ 4 h) led to the reduction of LAS removal ratio and high LAS_{Eff} due to continuously growth of symbiotic biofilm (Schlafer and Meyer, 2017). Consistent with previous findings that lower HRT resulted to high LAS loading rate, and biotoxicity of LAS led to the decreased reactor performance in the MBBfR in treating greywater (Yin et al., 2023).

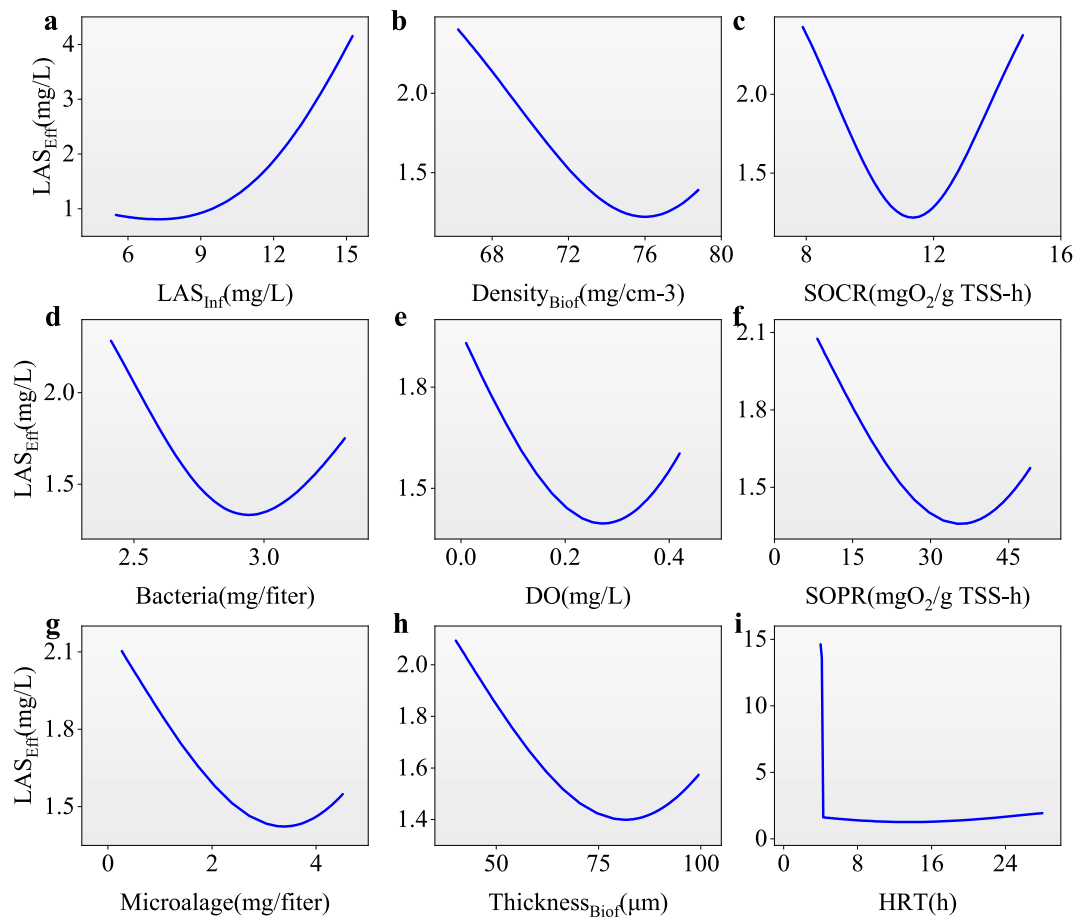


Fig. 7. The partial dependence plots between input variable (LAS_{Infl}, Density_{Biof}, SOCPR, Bacteria, DO, SOPR, Microalgae, Thickness_{Biof}, HRT) and the linear alkylbenzene sulfonate (LAS) mineralization in the microalgal-bacteria biofilm reactor. LAS_{Eff} and LAS_{Infl} are effluent and influent LAS concentrations, respectively; Density_{Biof} and Thickness_{Biof} are biofilm density and thickness, respectively; SOPR and SOCPR are specific oxygen production and consumption rates, respectively; HRT is hydraulic retention time and DO is dissolved oxygen.

3.3. Applicability evaluation of SVM model

PDP provides a visual depiction of the marginal effect of a variable on the response, quantifying this effect by altering the mean response (Biecek, 2018). PDP assumes independence between the feature for which it is calculated and the other features (Xin et al., 2022). According to the PDP, the effluent LAS is negatively correlated with biofilm density, indicating that the biofilm density positively contributed to LAS removal. In the initial region of MBBfR (see Supplementary material), with the increasing of bacterial concentration, LAS degradation efficiency declined, but high microalgae content led to improved LAS removal ratio, which should be contributed to the improved oxygen production during photosynthesis (Li et al., 2023; Schlafer and Meyer, 2017). Upon the biosorption of LAS in the symbiotic biofilm, increasing of both biofilm density and thickness led to the decrease of effluent LAS. DO and SOCPR exhibited inflexion points for LAS removal at 0.13 mg/L and 11.8 mg O₂/(g TSS•h), respectively. In this region, LAS exposure induced a toxic response in the symbiotic biofilm. As greywater flows into the third region of the MBBfR, low residual LAS led to the reduction of LAS biotoxicity. As shown in Fig. 7, according to the constructed ML model prediction outputs, MBBfR enabled the efficient removal of LAS at the optimal operation conditions of HRT 4.28 h, DO 0.25 mg/L, SOCPR and SOPR 11.4 and 35.2 mg O₂/(g TSS•h), biofilm density and thickness 76.0 mg/cm³ and 81.8 μm, bacteria and microalgae concentrations of 2.94 and 3.26 mg/fiter, respectively. Low HRT and DO with stable and efficient LAS removal ratio could significantly reduce the energy-requirement during greywater treatment, which favored the practical

application of MBBfR in greywater and other types of sewage treatment.

For checking the reliability of ML model, the same MBBfR was continuously operated for 20 days with HRT and DO of 4.28 h and 0.25 mg/L (same operation conditions of other parameters), respectively. Experimental results showed that effluent LAS concentration was maintained at 1.52 ± 0.12 mg/L. According to the ML model, the predicted effluent LAS concentration was 1.48 mg/L. Low relative errors (2.63 %) between the predict results and the experimental data indicated that the selected ML model could accurately predict LAS removal in the MBBfR.

3.4. Significance and environmental implications

Aeration-free MBBfR involving complex reactions can be applied for efficient LAS mineralization during greywater treatment. Modeling these intricate interactions using reaction mechanisms proves challenging. Consequently, a ML model was utilized for the simulating and prediction of effluent LAS levels using reactor operation conditions and biofilm physicochemical properties as input variables. After comparing various model algorithms, the optimal model is identified and applied for the investigation of MBBfR in LAS mineralization. The model elucidates the contributions of biofilm characteristics to effluent LAS and the partial dependence of different variables on LAS removal. Based on model interpretation, the operating conditions of the MBBfR were further adjusted for further enhancing LAS removal during greywater treatment.

LAS mineralization during greywater treatment is complex, and

understanding LAS removal flux and mechanisms by bacteria and microalgae in the symbiotic biofilm is challenging. The constructed ML model enabled the stimulation and prediction of LAS mineralization in the MBBfR. The model is thoroughly explained and the reaction process is clearly quantified. This work provides valuable data for future research on energy-efficient LAS mineralization during greywater treatment in the MBBfR. Addressing environmental challenges with machine learning represents a significant advancement. Therefore, user-friendly code is provided to facilitate the integration of greywater processing with artificial intelligence. Open data and source code make AI applications accessible to environmental researchers.

4. Conclusions

This study combines a specific case study of ML and MBBfR operation optimization. SVM methods can effectively and accurately predict LAS mineralization ($R^2 = 0.961$, RMSE = 0.251, MAE = 0.153 in the testing set). SHAP and PDP methods explained LAS removal drivers and provided transparency in data-driven modeling. Low-energy operating conditions (HRT = 4.28 h, DO = 0.25 mg/L) were optimized with significant energy saving potential. LAS_{inf} , $Density_{Biof}$ and SOCR had the greatest influence on LAS_{Eff} , and bacteria directly impact LAS mineralization. The findings bridge the research gap between ML models and MBBfR reactor design optimization.

CRedit authorship contribution statement

Libo Xia: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Beibei Wu:** Writing – review & editing, Visualization, Software, Data curation. **Xiaocai Cui:** Writing – review & editing, Visualization, Software. **Ting Ran:** Writing – review & editing, Visualization, Software. **Qian Li:** Writing – review & editing. **Yun Zhou:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biortech.2024.132028>.

Data availability

Data will be made available on request.

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